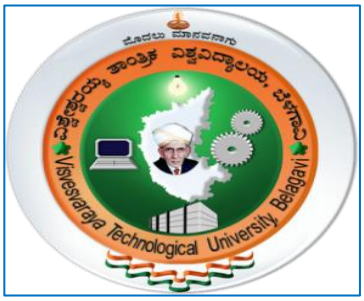
VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Machhe, Belagavi, Karnataka 590018



INTERNSHIP-I REPORT

On

"C++ PROGRAMMING AND WEB TECHNOLOGY"

Submitted in partial fulfillment of the requirement
for the award of the degree of

Bachelor of Engineering

In

Information Science and Engineering

By

KRITI P DESAI	**1BG25IS029**
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Vidyayāmruthamashnuthe

B.N.M. Institute of Technology

An Autonomous Institution under VTU, Approved by AICTE

Department of Information Science and Engineering

2026 – 2027

B.N.M. Institute of Technology

An Autonomous Institution under VTU, Approved by AICTE

DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

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CERTIFICATE

Certified that **KRITI P DESAI (1BG25IS029) TANMAYI KT (1BG25IS058) TEJAS GOWDA M (1BG25IS059)** has carried out the Internship I – 25ISE146 on **C++ Programming and Web Technology** conducted internally from **03rd August 2026 to 20th August 2026**. Candidate has fulfilled all the requirements prescribed in the curriculum. Candidate has incorporated all the corrections/suggestions indicated during the Internal Assessment. The Internship I – 25ISE146 work has been approved as it satisfies the curriculum requirements.

Dr. Laxmi V **Associate Professor** **Dept. of ISE, BNMIT**	**Dr. S Srividhya** **Professor & Head,** **Dept. of ISE, BNMIT**	**Dr. S Y Kulkarni** **Additional Director &Principal** **BNMIT**

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DECLARATION

KRITI P DESAI (1BG25IS029),TANMAYI KT (1BG25IS058) and TEJAS GOWDA M (1BG25IS059) a students of B.E. in Information Science and Engineering, Bengaluru, hereby declare that, I have carried out **15** days of Internship I – 25ISE146 at **BNM Institute of Technology, Post Box No.7087, 27th Cross, 12th Main, Banashankari 2nd Stage, Bengaluru-560070** From **03rd August 2026 to 20th August 2026**. under the guidance **of Dr. Laxmi V, Associate Professor, Department of ISE, BNMIT.** This Internship I – 25ISE146 report is submitted to Visvesvaraya Technological University, Belagavi in partial fulfillment of the requirement for the award of degree of Bachelor of Engineering in Information Science & Engineering during the academic year 2026-27.

Place: Bengaluru **Signature of Student**

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I would like to thank our Internship-I – 25ISE146 Coordinator **Dr. Laxmi V**, Associate Professor Department of Information Science and Engineering, BNMIT, for being the guiding force towards the successful completion of our Internship-I – 25ISE146.

KRITI P DESAI (1BG25IS029)
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ABOUT THE DEPARTMENT

The Department of Information Science & Engineering was established in the year 2001 with an intake of 60. The department has qualified and dedicated faculty members to provide quality education to students. The department is accredited by NBA for the academic year 2024-25 to 2027-28. The department has well-established R & D Centre recognized by VTU, Belagavi, and faculty are actively involved in the research activities, publishing their work in various Journals and Conferences in thrust areas like, Network Security, Internet of Things, Computer Vision, Machine Learning, Artificial Intelligence, Image Processing and Deep Learning etc. The department has student projects funded by IEDC. The Department has organized three online and one offline FDPs on thrust areas funded by AICTE Training and Learning Academy (ATAL). The department has state-of-the-art infrastructure to support research and develop students into professionals. Students are trained with multi-dimensional skill sets for their holistic growth. The department regularly organizes skill development programs, internships, workshops, alumni series, hackathon, technical talks and industrial visits to keep the students abreast with technology.

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CHAPTER 1

INTERNSHIP DESCRIPTION

1.1 Introduction to C++ Programming

C++ is a powerful, high-performance programming language widely used for system software, computational engines, and applications that demand efficient memory management and fast execution. During this internship, C++ was adopted as the core language for the timetable optimization engine because of its ability to handle intensive constraint checking and search algorithms with minimal runtime overhead.

Key concepts applied in the project include:

- Object-oriented programming (classes for Groups, Teachers, Subjects, Rooms and Sessions).
- Standard Template Library (STL) containers such as vectors, maps and sets for storing institutional data and intermediate solutions.
- File I/O for reading input.txt (prepared by the Java layer) and writing the optimized timetable to output.txt.
- Algorithm design — backtracking with forward checking, graph-coloring heuristics for seed generation, and a genetic algorithm for soft-constraint optimization.

The C++ engine performs the following major operations: loading groups, teachers, subjects and rooms; checking teacher availability and conflicts; checking group and room conflicts; verifying room capacity and type; ensuring required subject periods; generating possible solutions; calculating optimization scores; removing duplicates; and selecting the best valid timetable.

1.2 Introduction to Web Technology

Web technologies form the presentation and interaction layer of the Smart Timetable system. HTML5 provides the structural markup for forms and dashboards, CSS3 is used for responsive styling and layout, and JavaScript enables client-side interactivity such as form validation and dynamic display of generated timetables.

During the internship the following web-technology skills were applied:

- Designing administrator dashboards and data-entry forms for groups, teachers, subjects and rooms.
- Presenting generated timetable results in a clear, tabular format that administrators can review and export.
- Integrating the front-end with the Java/JDBC back-end so that institutional data flows seamlessly from the user interface into the MySQL database and subsequently into the C++ optimization engine.

The combination of a modern web front-end with a high-performance C++ solver demonstrates a practical full-stack approach to solving a real institutional scheduling problem.

CHAPTER 2

LEARNING OBJECTIVES / INTERNSHIP OBJECTIVES

The primary learning and internship objectives of this Internal Internship (25ISE146) are:

- To gain practical proficiency in C++ programming, focusing on data structures, file handling, and algorithm implementation for constraint satisfaction and optimization.

- To acquire hands-on experience with web technologies (HTML, CSS, JavaScript) for building interactive administrative interfaces.

- To understand the integration of a multi-language stack (Java/JDBC, MySQL, C++, and web front-end) through file-based and database-mediated communication.

- To model a real-world administrative problem (academic timetable generation) as a Constraint Satisfaction Problem and apply classical AI techniques such as backtracking, graph coloring and genetic algorithms.

- To design and implement a usable administrator dashboard that allows data entry, constraint configuration, timetable generation and result visualization.

- To design and implement a well-structured MySQL database schema for managing academic entities (courses, faculty, rooms, time slots, student groups, etc.) and to ensure reliable data persistence and retrieval.

- To incorporate and handle realistic institutional constraints such as faculty availability, room capacity, course conflicts, consecutive lecture preferences, and workload balancing.

- To evaluate and compare the performance of different algorithms (backtracking, graph coloring, and genetic algorithms) with respect to solution quality, computational efficiency, and scalability.

- To develop systematic testing, debugging and validation strategies to verify the correctness, completeness and robustness of the generated timetables.

- To apply software engineering practices including modular design, version control, iterative development and documentation throughout the project lifecycle.

- To document the complete development process, including system requirements, implementation details, results, performance analysis and lessons learned, in the form of a formal internship report.

• To present and demonstrate the working system, incorporating feedback for continuous improvement.

CHAPTER 3

WEEKELY OVERVIEW OF INTERNSHIP ACTIVITIES

Table 1: Internship Activity Summary

	DATE	DAY	NAME OF THE TOPIC/ MODULE COMPLETED
1st WEEK	**03/08/2026**	**Monday**	Basic Syntax , Comments In C++ , Data Types , Variable Types , Variable Scope , Constants/Literals , Modifier Types , Operators , Arrays , Strings
	04/08/2026	**Tuesday**	Loop, Decision Making Statements And Functions, Pointers, References, Basic Input/Output, Data Structures,
	05/08/2026	**Wednesday**	Classes And Objects , Inheritance, Overloading (Operator)
	06/08/2026	**Thursday**	Overloading Function Polymorphism, Data Abstraction
	07/08/2026	**Friday**	Dynamic Memory , Namespaces ,
	08/08/2026	**Saturday**	Templates , C++ Stl
2nd WEEK	**10/08/2026**	**Monday**	Web Development Technologies, Basics, Web Development Prerequisites, Important Terminologies, Why Learn Web Development? Introduction to HTML- Features, editors, Comments, HTML Fundamental Concepts- Structure, Elements, Attributes, Fundamental Concepts- Semantics, Tags HTML projects- • Simple Portfolio Website • Design an Event Web Page • Beginner HTML Project Ideas
	11/08/2026	**Tuesday**	Introduction to HTML Tables Practical Session on HTML Tables

			Programs using Tables
	12/08/2026	**Wednesday**	Lists- ordered, unordered, description list Table Formatting Forms, Fundamentals of CSS
	13/08/2026	**Thursday**	Javascript Document Object Module, Client and Server side Programs Programs using forms and CSS
	14/08/2026	**Friday**	Introduction to ODBC/JDBC ODBC/JDBC Demo Program on Client server communication
3rd WEEK	**17/08/2026**	**Monday**	Project Work
	18/08/2026	**Tuesday**	
	19/08/2026	**Wednesday**	Demo Day
	20/08/2026	**Thursday**	

CHAPTER 4

SYSTEM REQUIREMENT SPECIFICATION

4.1 User Requirements

The system is intended for use by academic administrators and timetable coordinators. The principal user requirements are:

- Ability to enter and manage institutional data — student groups/batches, teachers/faculty, subjects/courses and rooms/laboratories.
- Configuration of working days, periods per day and subject-hour requirements.
- One-click generation of a conflict-free timetable that respects hard constraints (no faculty/room/batch clashes).
- Presentation of the generated timetable together with an optimization score so that the administrator can assess quality.
- Clear diagnostic messages when the supplied constraints are infeasible.
- Export or display of the final schedule for publication and further manual adjustment if required.

4.2 Software Requirements

1. Programming Languages: C++ (optimization engine), Java (application & database layer), HTML, CSS and JavaScript (front-end).
2. Database: MySQL — stores groups, teachers, subjects and rooms.
3. Database Connectivity: JDBC (Java Database Connectivity).
4. Data Access Layer: DAO classes (Group DAO, Teacher DAO, Subject DAO, Room DAO).
5. File-Based Communication: input.txt (data for C++ engine) and output.txt (generated result).
6. IDE: Visual Studio Code.
7. Operating System: Windows / Linux.

4.3 Hardware Requirements

- Processor: Intel i5 (or equivalent) or higher.
- RAM: 8 GB minimum (16 GB recommended for large institutional datasets).
- Storage: 5 GB free disk space.

CHAPTER 5

IMPLEMENTATION AND RESULTS DISCUSSION

Tasks Executed

The implementation followed a layered architecture that cleanly separates the web front-end, the Java/JDBC data layer, the MySQL database and the C++ optimization engine. The principal tasks executed during the internship are summarized below.

5.1 C++

The C++ component constitutes the computational heart of the system. The main implementation tasks were:

- Design of data structures to represent groups, teachers, subjects, rooms and teaching sessions.
- Parsing of institutional data from input.txt prepared by the Java layer.
- Hard-constraint checking: teacher availability and uniqueness, group uniqueness, room uniqueness, capacity and type, and required weekly periods for each subject–batch pair.
- Constructive seeding via a DSATUR-style graph-coloring heuristic to obtain an initial conflict-free timetable.
- Genetic-algorithm improvement phase: tournament selection, constraint-preserving crossover, low-probability mutation, and steady-state replacement driven by a weighted soft-constraint fitness function (gap minimization, workload balance, preferred slots).
- Removal of duplicate solutions and selection of the best-scoring valid timetable, written to output.txt.

Testing on representative department-level datasets confirmed that the engine produces fully conflict-free timetables and that the optimization score improves across generations until a plateau or generation limit is reached.

5.2 Web Technology

The web front-end was developed using HTML5, CSS3 and JavaScript. Key implementation tasks included:

- Student Group Management Interface — forms for creating and editing batches.

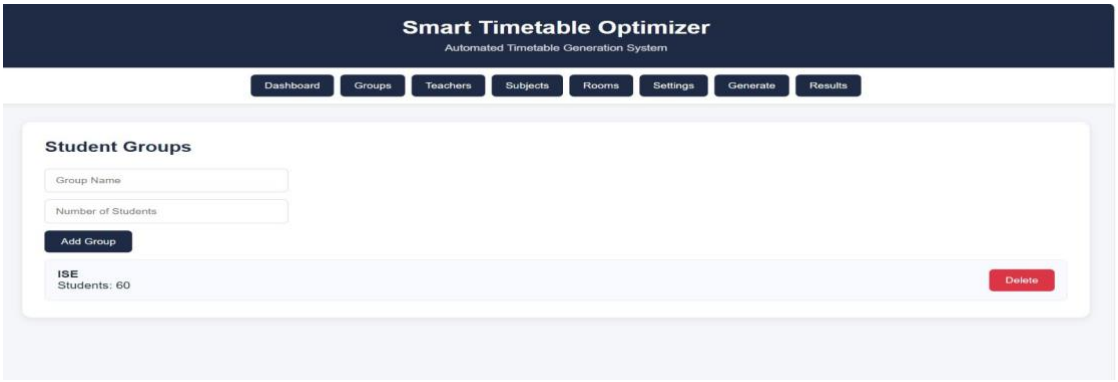


Figure 5.1: Student groups management interface

- Administrator Dashboard providing navigation to all data-entry modules and the generation trigger.



Figure 5.2: Administrator dashboard showing summary counts of institutional data

Figure 5.1: The Student Groups page displays a simple form with fields for Group Name and Number of Students, along with an “Add Group” button. Below the form, the existing group “ISE” with 60 students is listed, each entry having a red Delete button.

Figure 5.2: The dashboard shows four summary cards displaying the current institutional data counts: Groups – 1, Teachers – 8, Subjects – 8, and Rooms – 7. The top navigation bar provides quick access to all modules.

- Institutional Data Management screens for teachers, subjects and rooms.



Figure 5.3: Teachers management interface



Figure 5.4: Subjects management interface

Figure 5.3: The Teachers page contains a form to enter Teacher Name, Subject, and Availability. The list below shows teachers such as Rajesh (DSA, All Periods), Ananya (DBMS, All Periods), and Vikram (Computer Networks, All Periods), each with a Delete option.

Figure 5.4: The Subjects page features a form for Subject Name, Teacher, Periods, Group, and Room Type. Existing subjects listed include DSA (Teacher: Rajesh, Group: ISE, Periods: 4, Room Type: Classroom) and DBMS (Teacher: Ananya, Group: ISE, Periods: 4).

- Constraint Configuration Interface allowing the administrator to set working days, periods and soft-constraint weights.

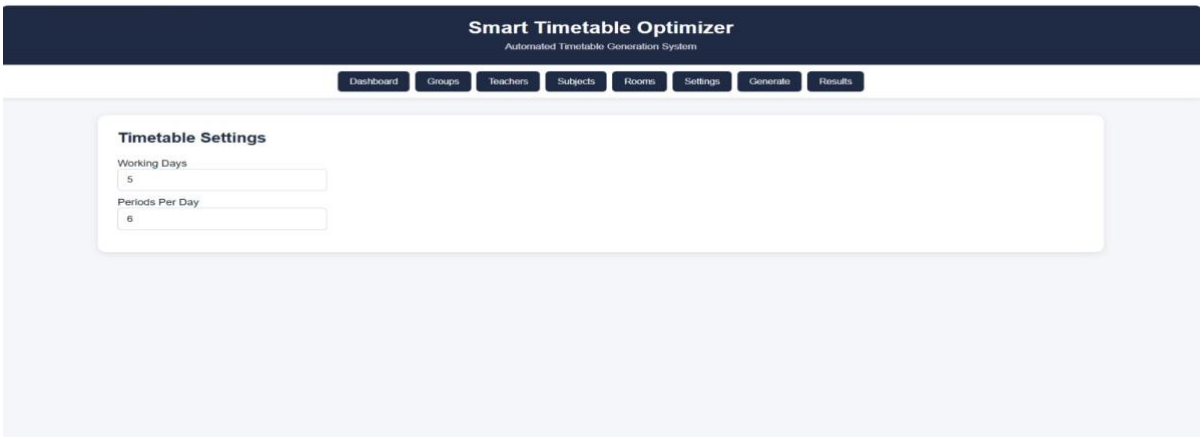


Figure 5.5: Timetable Setting/Constraint Configuration screen

- Result Display Module that reads the optimized timetable from the application layer and presents it in a readable grid together with optimization statistics.

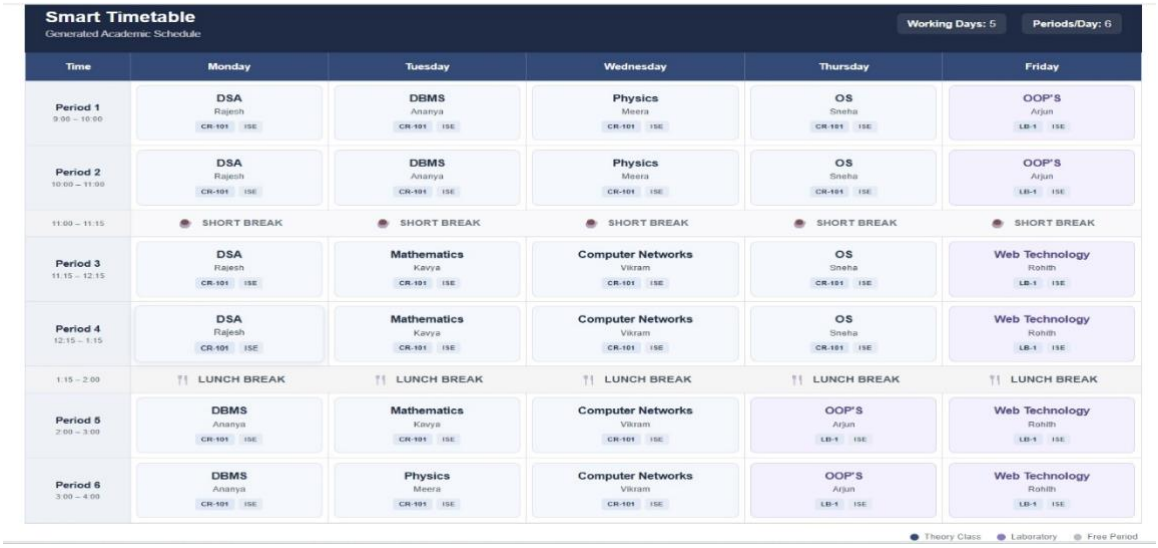


Figure 5.6: Sample Generated conflict-free academic timetable

Figure 5.5: The Settings page is minimal and clean, allowing the administrator to configure Working Days (set to 5) and Periods Per Day (set to 6).

Figure 5.6: The generated timetable presents a complete Monday-to-Friday grid with six periods, short breaks, and a lunch break. Each cell displays the subject, teacher name, room (CR-101 or LB-1), and group (ISE), with clear visual distinction between theory and laboratory classes.

The front-end communicates with the Java application, which in turn uses JDBC and DAO classes to persist data in MySQL. When generation is requested, the Java layer prepares input.txt, invokes the C++ engine, reads output.txt and returns the result to the browser for display.

Overall, the integrated system successfully generates valid, optimized timetables and presents them through a clean administrative interface, fulfilling the functional goals of the internship project.

5.3 Tools and Technologies Used

The following tools and technologies were used in the development of the Smart Timetable Possibility & Optimization Engine:

- Frontend: HTML, CSS and JavaScript were used to design the user interface, administrator dashboard, data-entry forms and for displaying the generated timetable.

- Backend / Application Layer: Java was used to implement the application logic and handle communication between the frontend and the database.

- Database Connectivity: JDBC (Java Database Connectivity) was used to establish the connection between the Java backend and the MySQL database.

- Database: MySQL was used to store all timetable-related data such as groups, teachers, subjects, rooms and constraints. SQL was used for performing database operations (insert, update, delete and retrieve).

- Optimization Engine: C++ was used to develop the high-performance optimization engine. The C++ program generates a conflict-free and optimized timetable by considering different hard and soft constraints.

The frontend interacts with the Java backend, which uses JDBC to communicate with the MySQL database. The optimized timetable generated by the C++ engine is then retrieved and displayed on the frontend.

5.4 Detailed Project Task

The system allows an administrator to:

1. Enter and manage institutional data (Student Groups, Teachers, Subjects, and Rooms) through a web-based interface.

2. Configure working days, periods, and constraints.

3. Generate a complete academic timetable with a single click.

4. View the generated conflict-free timetable along with an optimization score.

5. Ensure that the timetable satisfies all hard constraints (no teacher, room, or group clashes) and optimizes soft constraints (workload balance, reduced gaps, preferred time slots, etc.).

Technology Stack Used:

- Frontend: HTML, CSS, JavaScript
- Backend: Java
- Database Connectivity: JDBC
- Database: MySQL (with SQL for operations)
- Optimization Engine: C++

CHAPTER 6 – CONCLUSION

6.1 Overview of Internship Experience

The Internal Internship provided a valuable opportunity to apply classroom knowledge of C++ and web technologies to a concrete, real-world problem. Working on the Smart Timetable Possibility & Optimization Engine deepened understanding of constraint satisfaction, evolutionary algorithms, full-stack integration and the practical challenges of building software that must be usable by non-technical administrators. Collaboration among team members, regular guidance from the faculty mentor, and iterative testing against realistic datasets strengthened both technical skills and professional work habits.

6.2 Challenges Faced

Several challenges were encountered and overcome during the internship:

- Designing a chromosome representation and repair operators that preserve hard constraints during genetic-algorithm crossover and mutation.
- Managing the file-based hand-off between the Java layer and the C++ engine while ensuring data consistency and clear error reporting.
- Balancing soft-constraint weights so that the optimization score produces schedules that are both mathematically good and practically acceptable to faculty.
- Creating an intuitive web interface that remains usable even when the underlying constraint set is complex.

These challenges were addressed through incremental development, systematic debugging, and continuous feedback from the project guide. The resulting system demonstrates that classical AI search techniques combined with a modern web front-end can deliver tangible operational benefits for educational administration.

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APPENDIX – I

SUSTAINABLE DEVELOPMENT GOALS

The Smart Timetable Possibility & Optimization Engine aligns with multiple United Nations Sustainable Development Goals (SDGs) by leveraging Artificial Intelligence and optimization techniques to improve educational administration. The project supports SDG 4 (Quality Education), SDG 5 (Gender Equality) and SDG 9 (Industry, Innovation and Infrastructure) by providing a sustainable, efficient and technology-driven platform for academic scheduling.

SDG 4: Quality Education

The system enhances the quality of education delivery by ensuring that academic timetables are generated efficiently, conflict-free and optimized for faculty workload and student convenience. By automating a complex administrative process, it frees up institutional resources that can be redirected towards teaching and learning. Students and faculty benefit from well-balanced schedules that reduce idle gaps and overcrowding, contributing to a more effective learning environment.

SDG 5: Gender Equality

The automated and transparent nature of the timetable generation process helps eliminate unconscious bias that may arise in manual scheduling. Preferred time slots, workload balance and room allocation are handled algorithmically based on objective constraints rather than subjective decisions. This promotes equitable distribution of teaching loads and access to resources for all faculty members irrespective of gender, supporting a fairer academic workplace.

SDG 9: Industry, Innovation and Infrastructure

By integrating classical constraint satisfaction techniques with genetic algorithms and a modern web-based architecture, the project demonstrates how AI-driven innovation can strengthen educational infrastructure. The system provides a scalable, reusable technological solution that institutions can adopt to modernize administrative processes, reduce manual effort and improve operational efficiency — contributing to resilient and innovative institutional infrastructure.